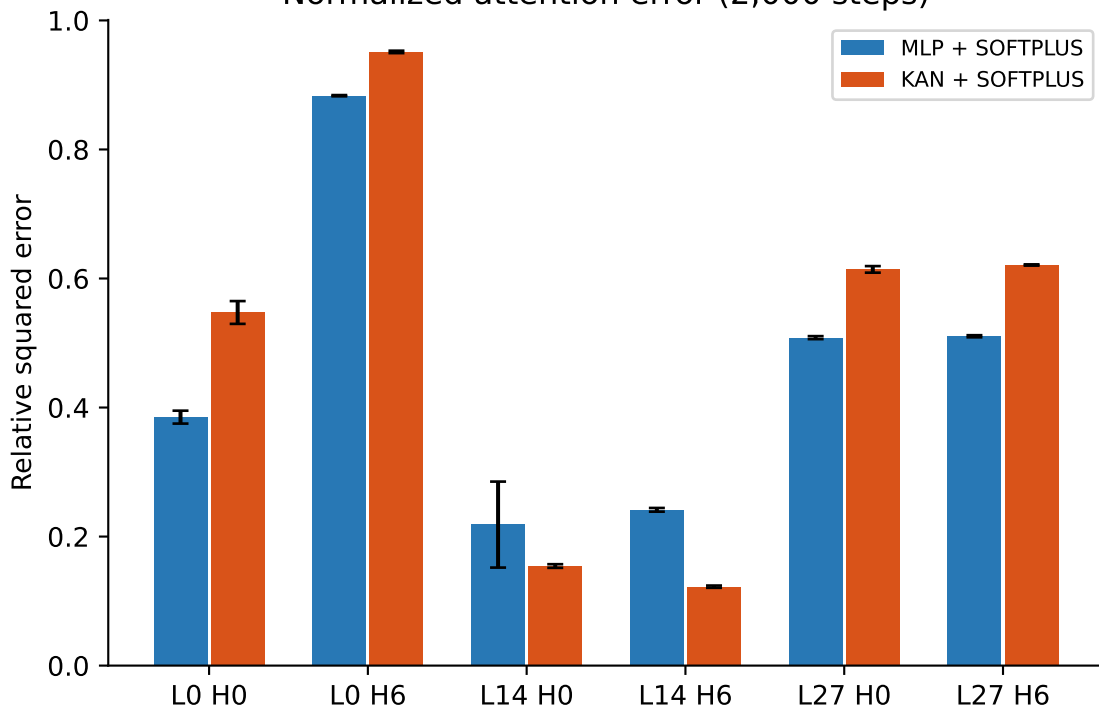
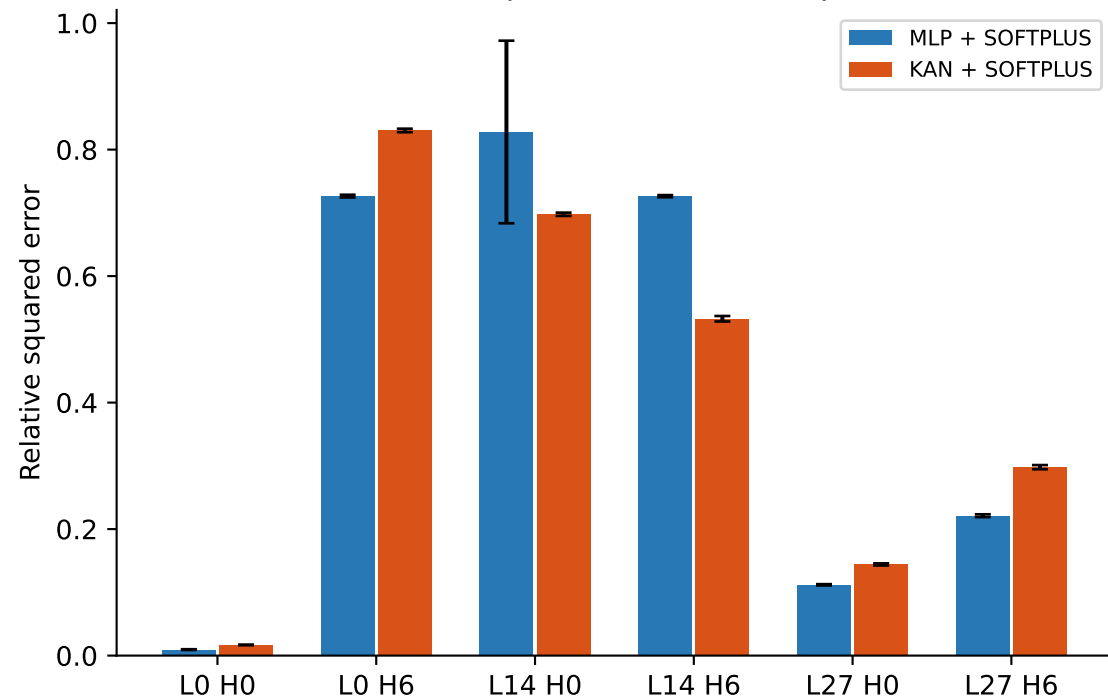


Frozen Qwen2.5-1.5B: independent Q/K maps, m=64, 3 seeds

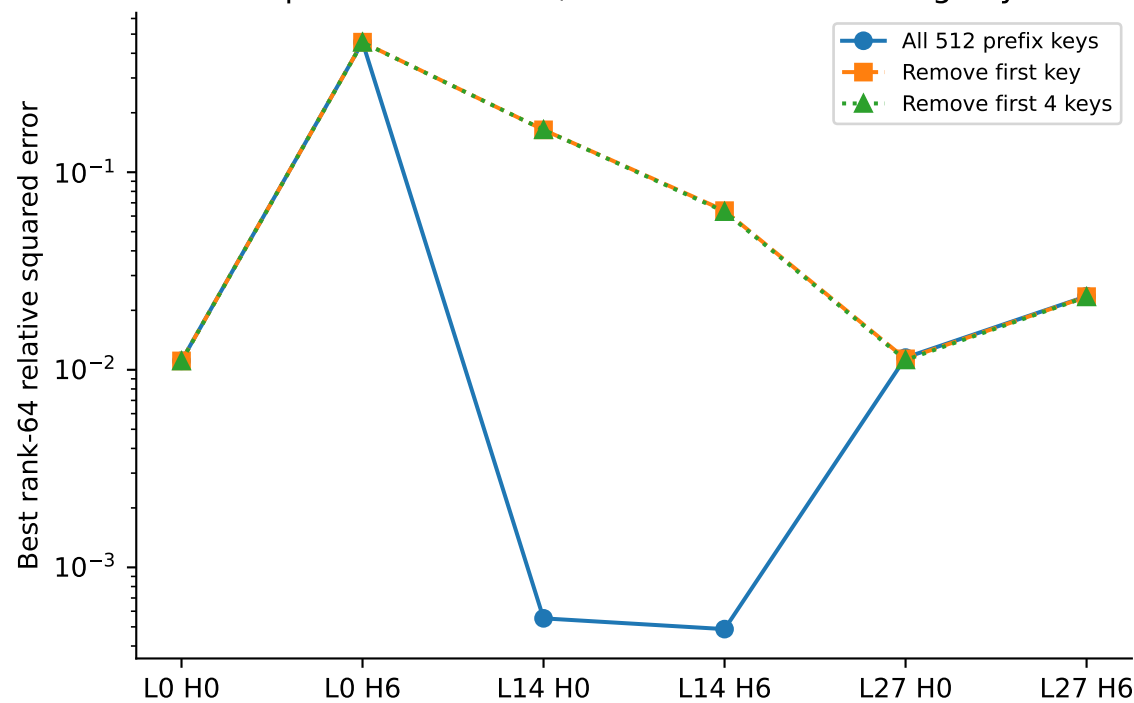
Normalized attention error (2,000 steps)



Head output error (2,000 steps)



Empirical SVD floors; re-normalize remaining keys



Normalized attention error (10,000 steps)

